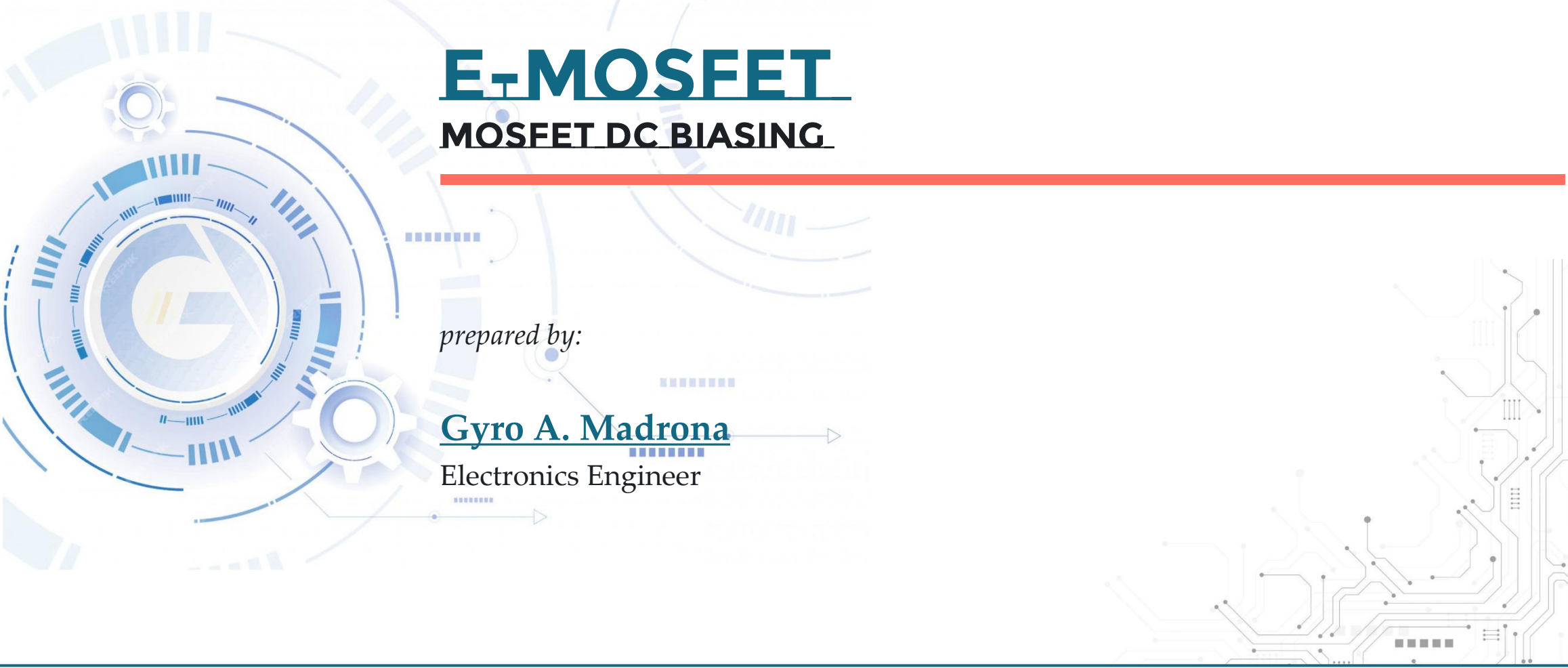




E-MOSFET

MOSFET DC BIASING

prepared by:

Gyro A. Madrona
Electronics Engineer

TOPIC OUTLINE

E-MOSFET Construction

Regions of Operation

E-MOSFET DC Biasing



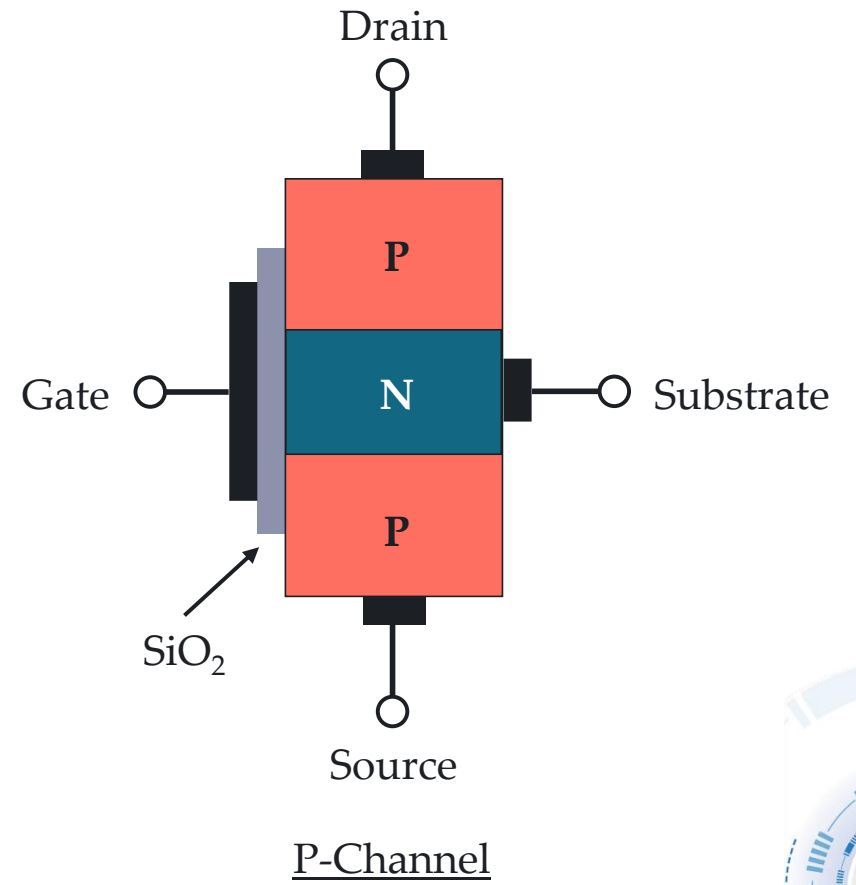
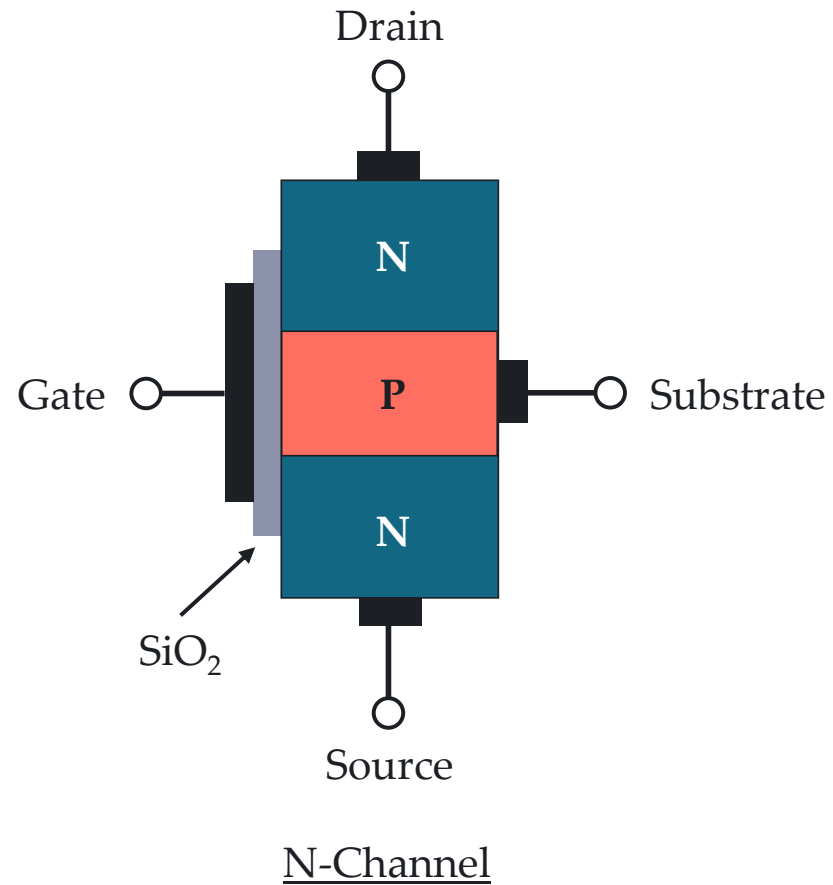
E-MOSFET CONSTRUCTION



CONSTRUCTION

E-MOSFET

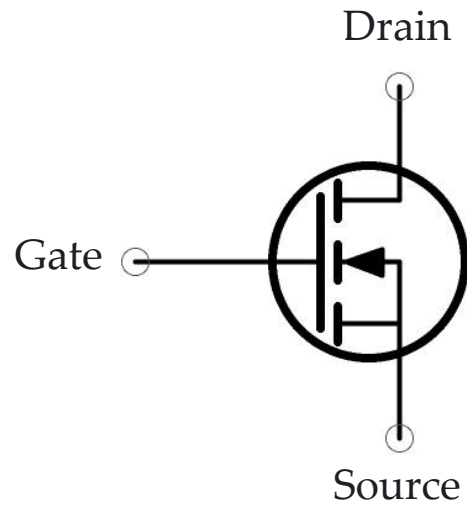
Enhancement Mode Metal-Oxide Semiconductor Field-Effect Transistor



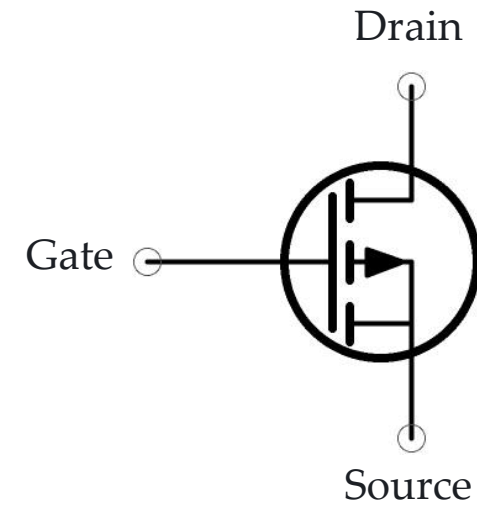
SCHEMATIC SYMBOL

E-MOSFET

Enhancement Mode Metal-Oxide Semiconductor Field-Effect Transistor



N-Channel

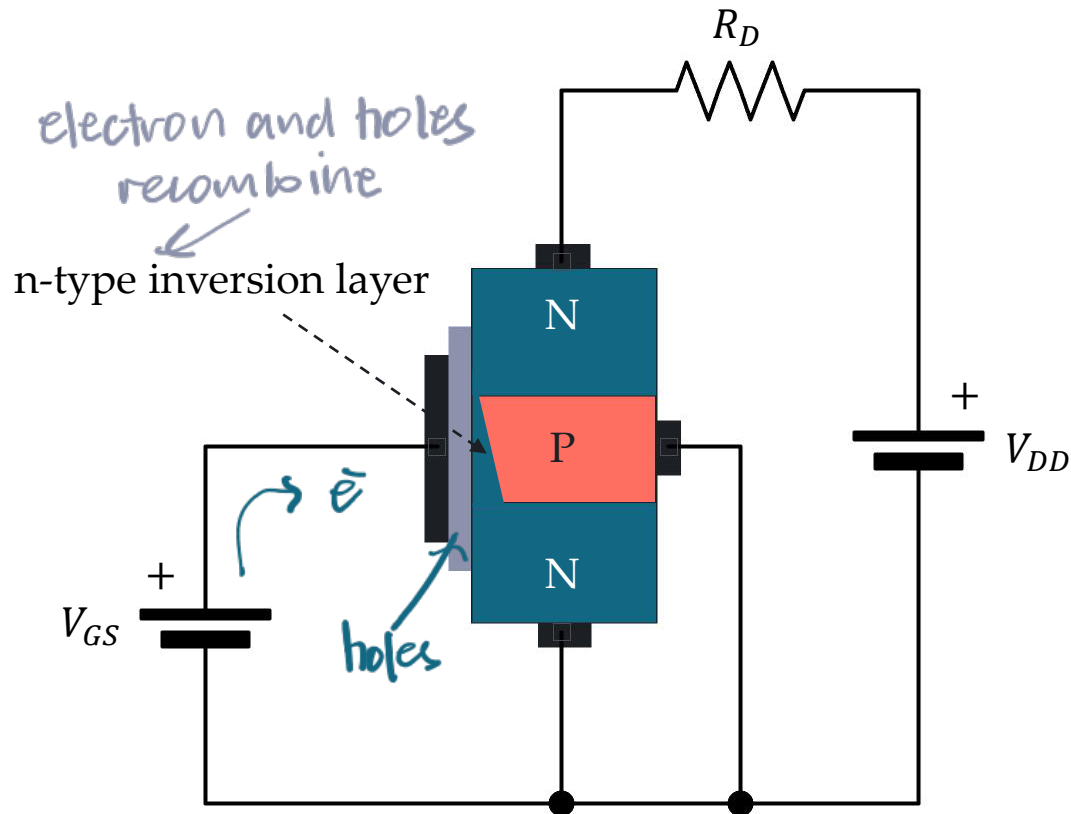


P-Channel

REGIONS OF OPERATION



N-TYPE INVERSION LAYER



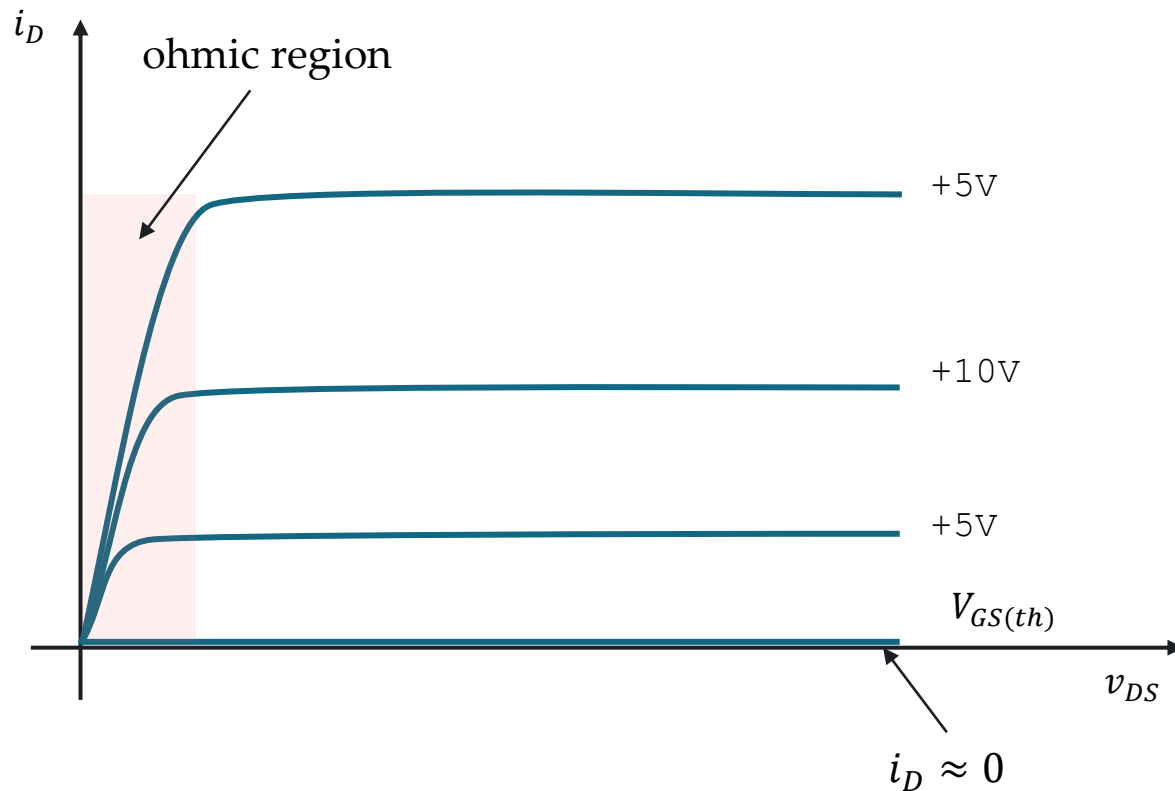
E-MOSFET is normally off

N-type inversion layer

A thin conducting layer near the silicon dioxide.

The **minimum voltage** that creates the n-type inversion layer is called the **threshold voltage** $V_{GS(th)}$ (1 to 3V for small-signal devices).

DRAIN CURVE



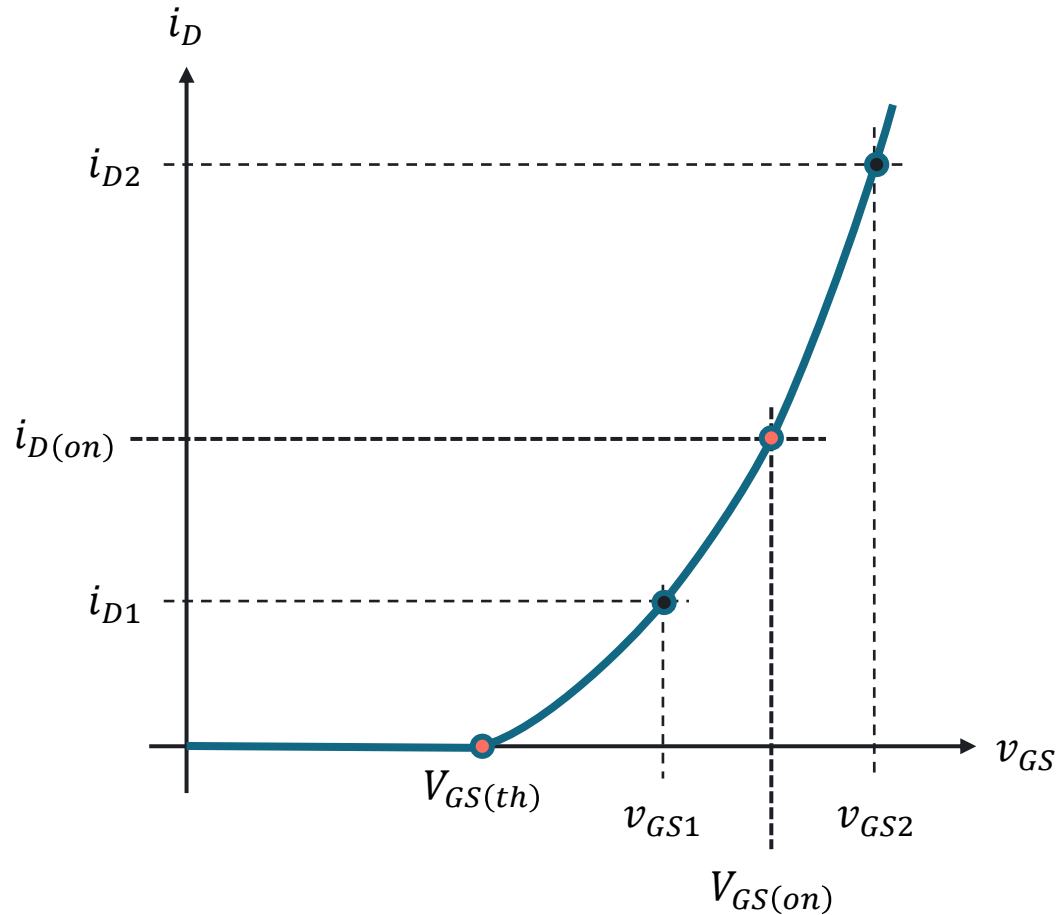
E-MOSFET is mainly use to operate in the **ohmic region** (switching device).

When v_{GS} is less than $v_{GS(th)}$, the drain current is approximately zero.



TRANSCONDUCTANCE CURVE

$i_{D(on)}$ and $V_{GS(on)}$ are given in the datasheet



Drain current

$$i_D = k(v_{GS} - V_{GS(th)})^2$$

$$(v_{GS} - V_{GS(th)})^2$$

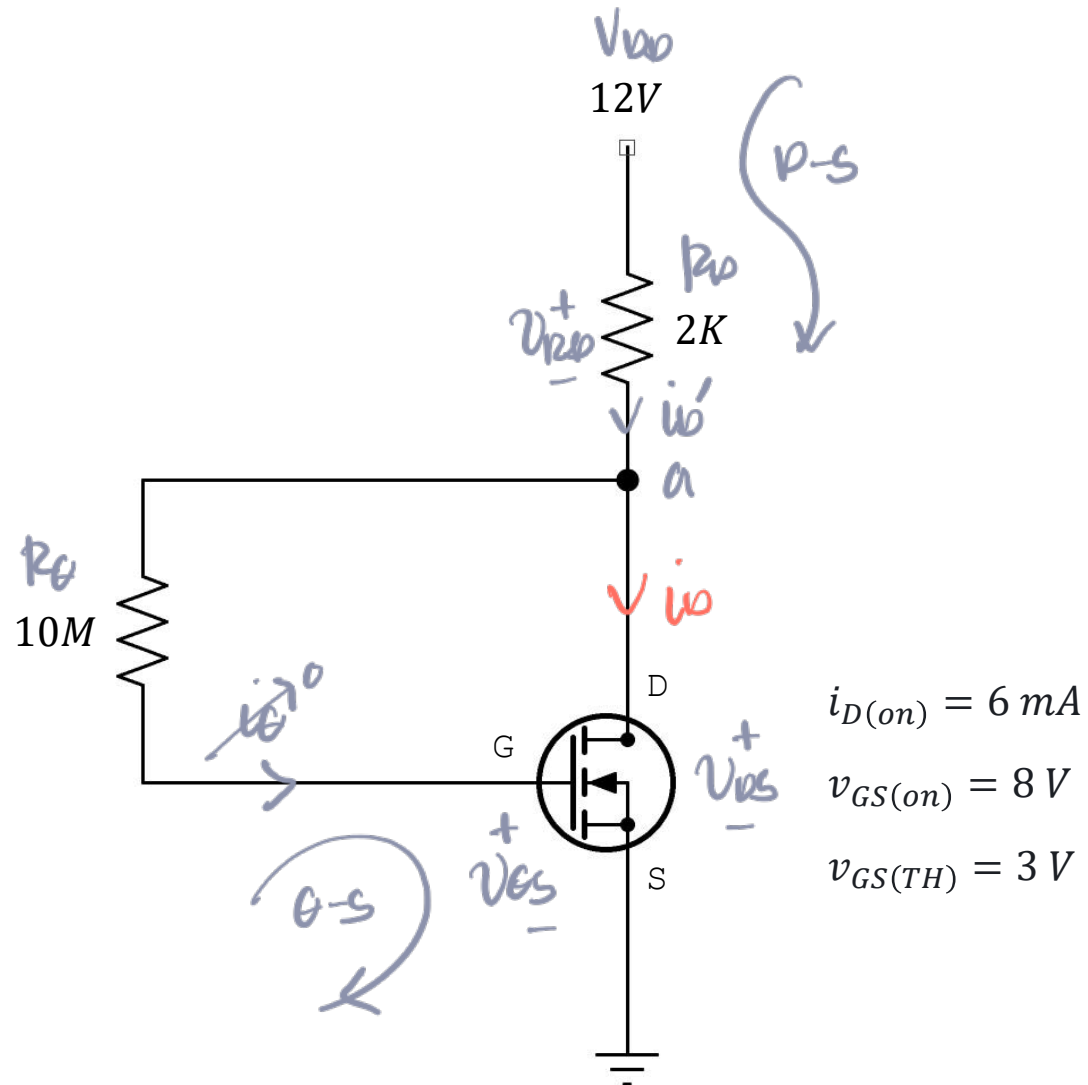
$$k = \frac{i_{D(on)}}{(v_{GS(on)} - V_{GS(th)})^2}$$



E-MOSFET DC BIASING



EXERCISE



For the given drain-feedback biasing arrangement, determine i_{DQ} and v_{DSQ} for the enhancement-type MOSFET. Sketch the transfer characteristics for $v_{GS} = 5V$ and $v_{GS} = 10V$.

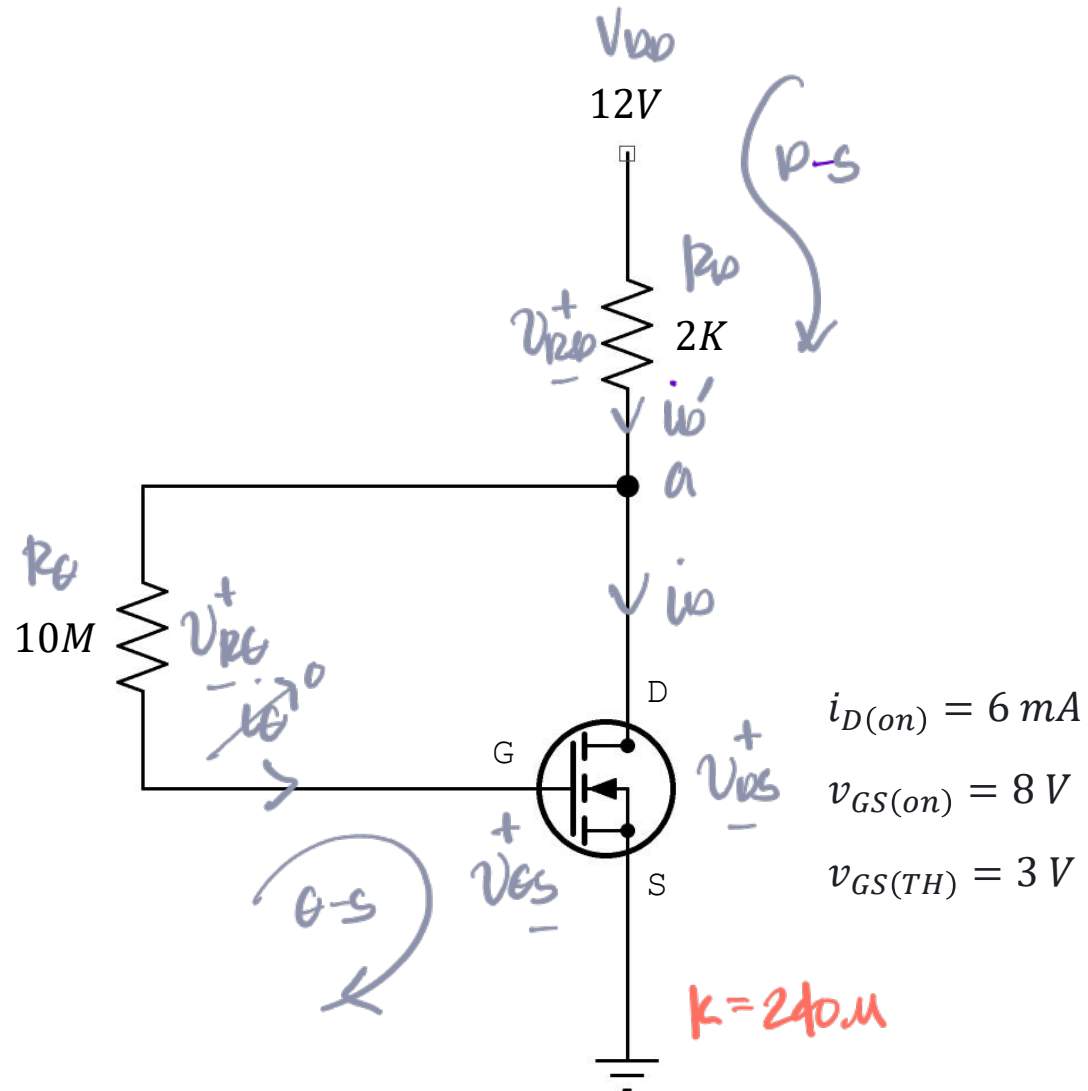
KCL @ a

$$-i_{D'} + i_{DQ} + i_G = 0$$

$$\underline{i_{DQ} = i_{D'}}$$



EXERCISE



Solution

Constant, k

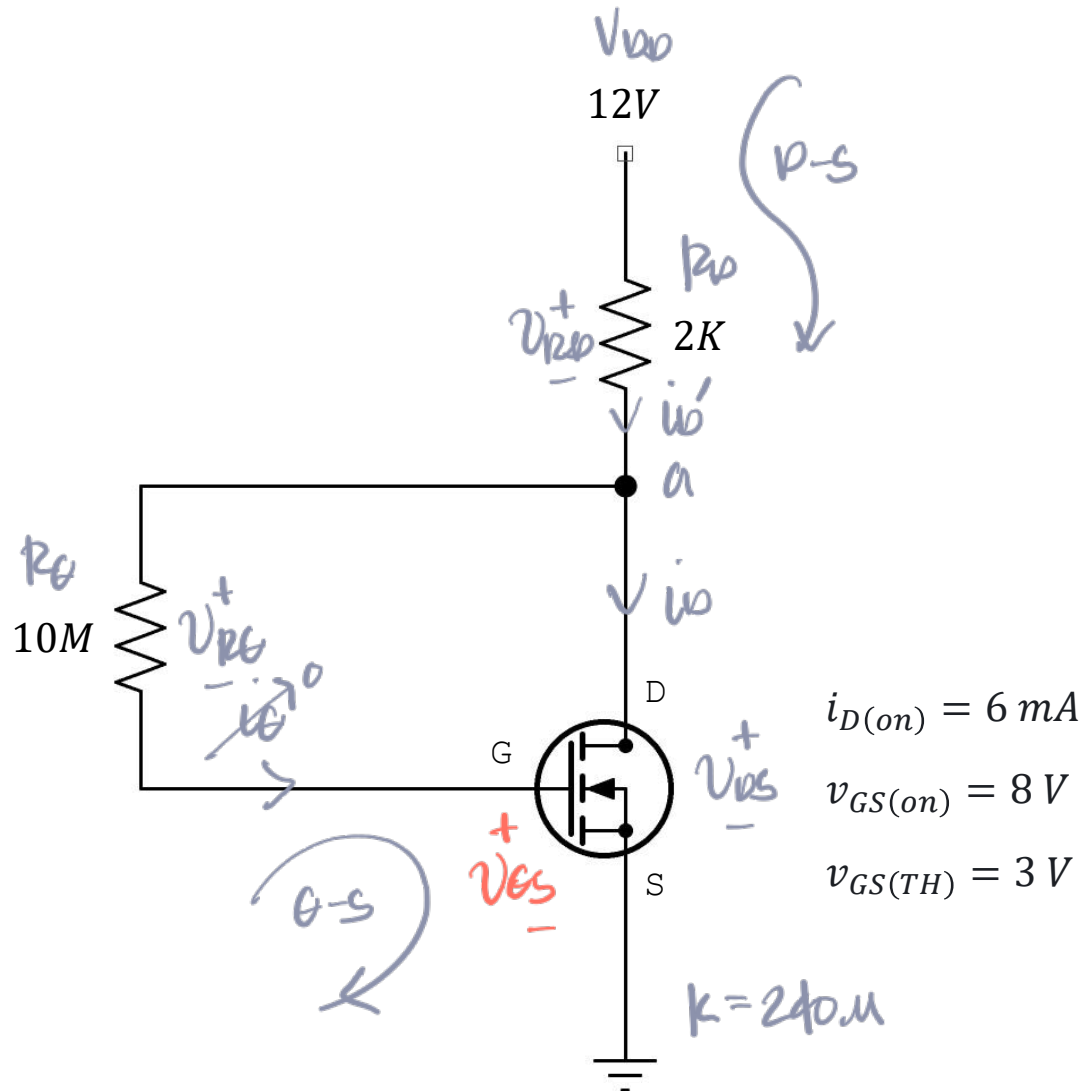
$$k = \frac{i_{D(on)}}{(v_{GS(on)} - v_{GS(th)})^2}$$

$$k = \frac{6 \text{ mA}}{(8 - 3)^2}$$

$$\underline{k = 240 \mu\text{A/V}^2}$$



EXERCISE



Solution

KVL @ θ-s

$$-V_{DD} + v_{RD} + \cancel{v_{RG}} + v_{GS} = 0$$

$$v_{GS} = V_{DD} - v_{RD}$$

$$v_{GS} = V_{DD} - i_D R_D$$

$$v_{GS} = V_{DD} - k(v_{GS} - v_{GS(TH)})^2 R_D$$

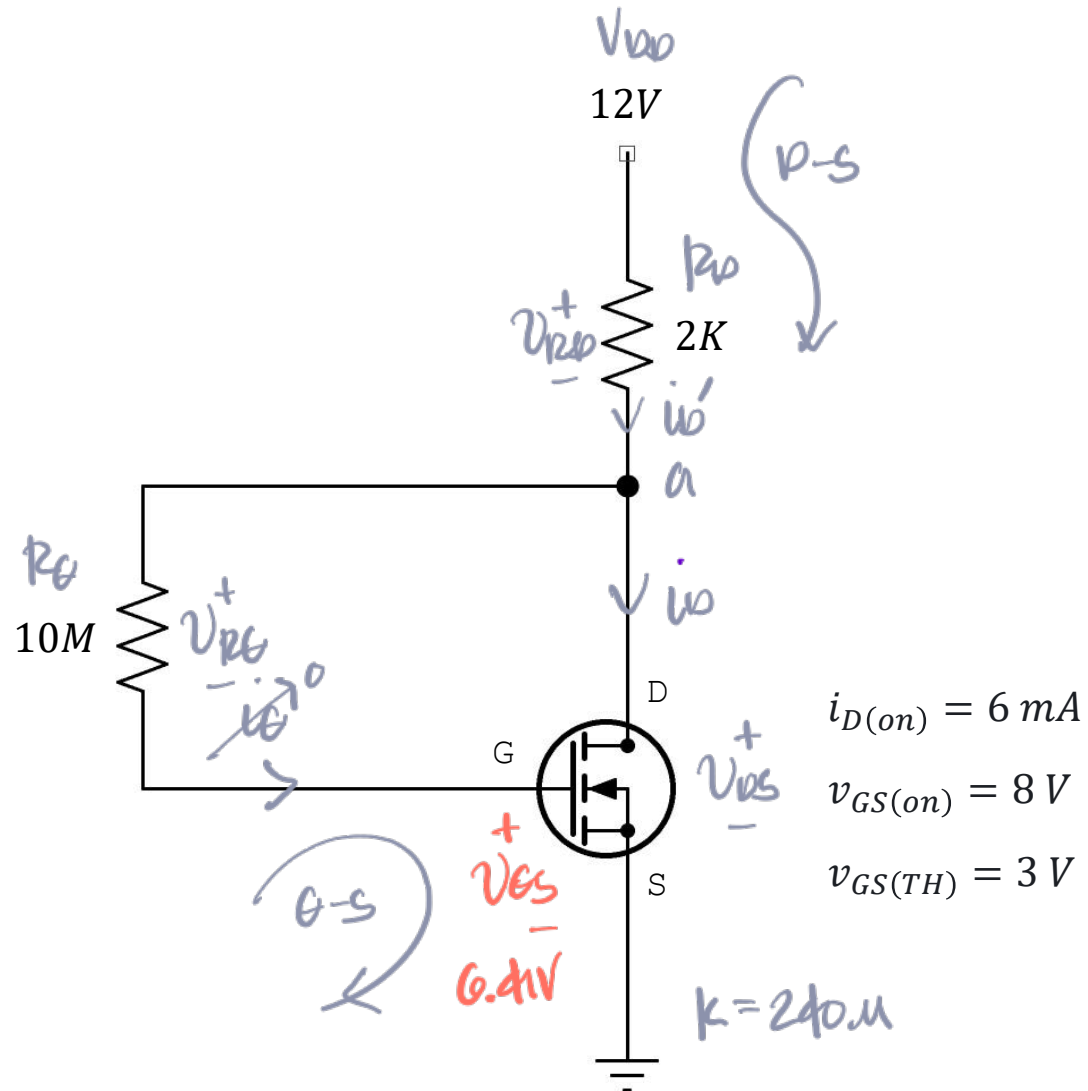
$$v_{GS} = 12 - 240\mu\text{A/V}^2 (v_{GS} - 3)^2 2K$$

$$v_{GS} = 12 - 0.48 (v_{GS}^2 - 6v_{GS} + 9)$$

$$v_{GS} = 12 - 0.48 v_{GS}^2 + 2.88 v_{GS} - 4.32$$

$-v_{GS}$

EXERCISE



Solution

KVL @ θ-s

$$0 = -0.48 v_{GS}^2 + 1.88 v_{GS} + 7.68$$

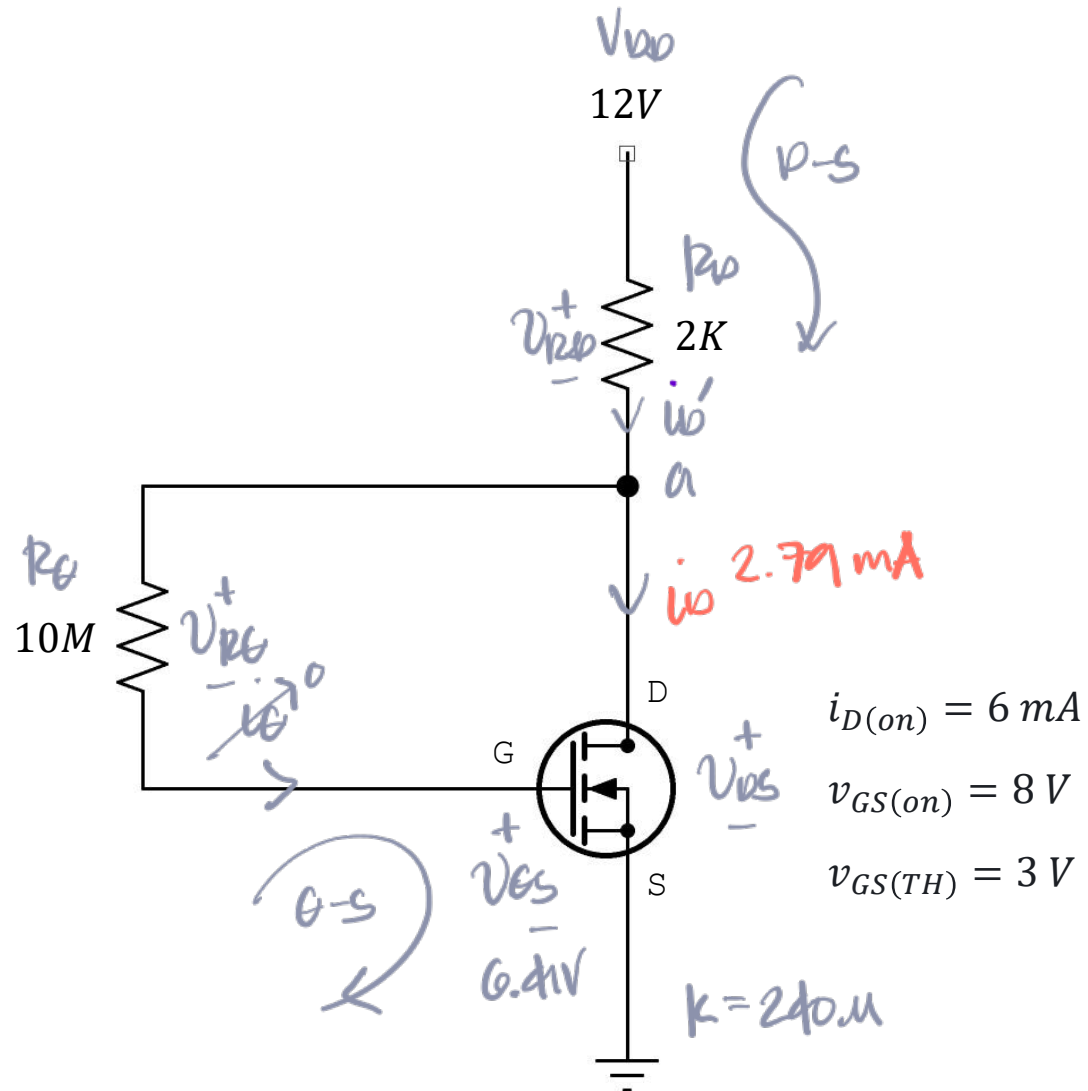
$$v_{GSQ} = 6.41 \text{ V}$$

ans

$$v_{GS} = -2.50 \text{ V}$$



EXERCISE



Solution

Drain Current

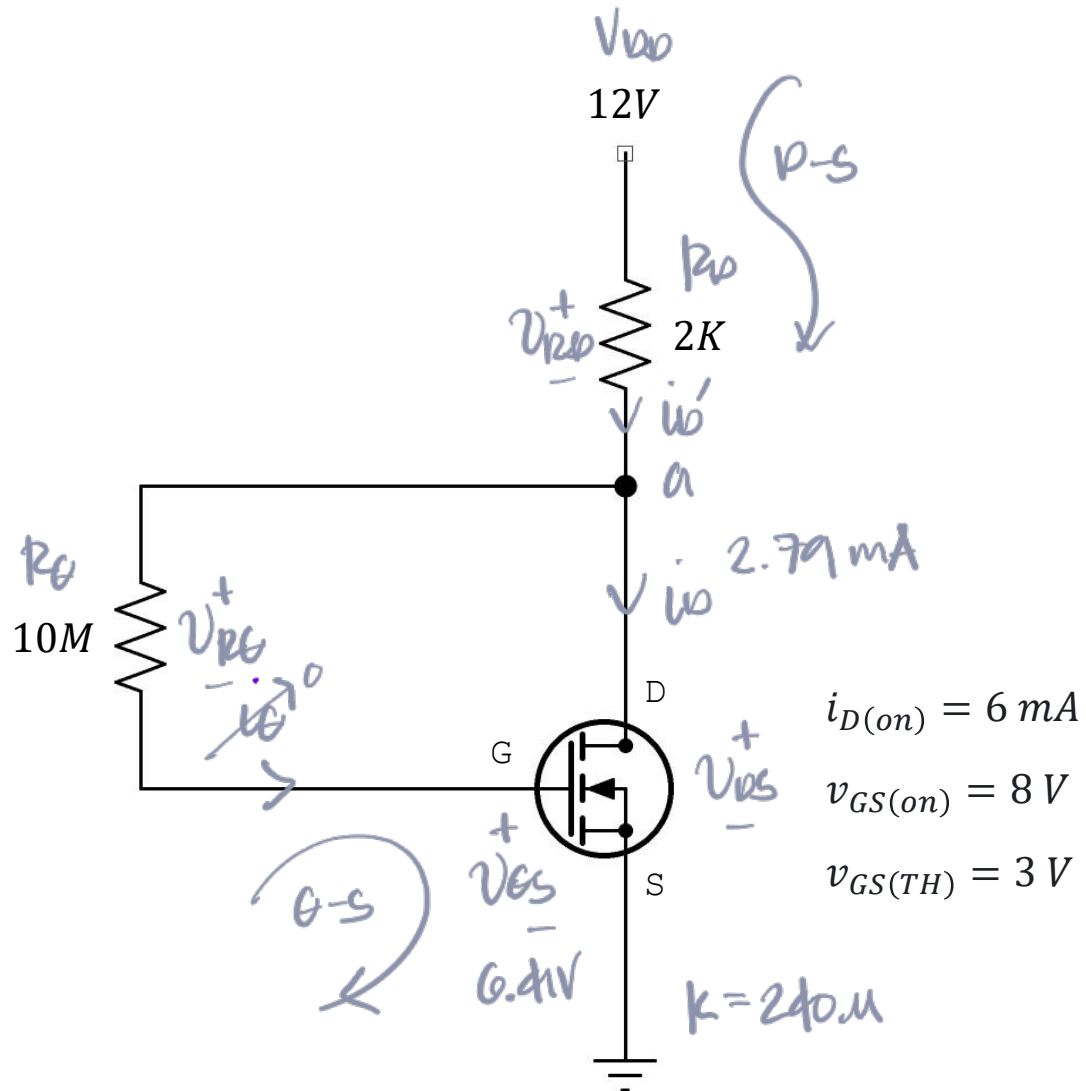
$$i_D = K (V_{GS} - V_{GS(th)})^2$$

$$i_D = 240\mu (6.41 - 3)^2$$

$$i_{DQ} = 2.79mA$$

ans

EXERCISE



Solution

Transfer characteristic

Let $v_{GS} = 5V$ ← between $v_{GS(on)} = 8V$ and $v_{GS(th)} = 3V$

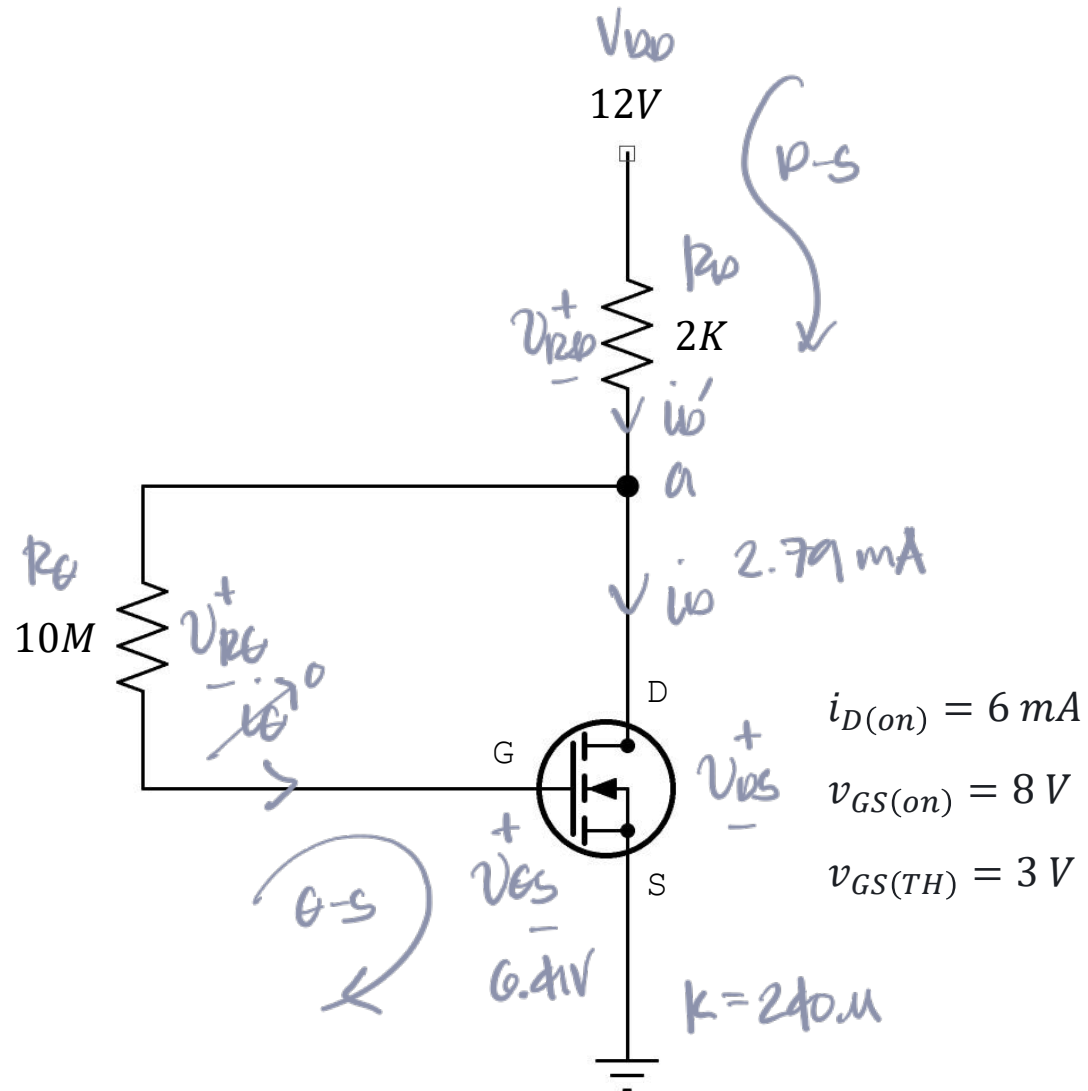
$$i_D = K(v_{GS} - v_{GS(th)})^2$$

$$i_D = 240 \mu (5 - 3)^2$$

$$i_D = 960 \mu A \quad | \quad v_{GS} = 5V$$



EXERCISE



Solution

Transfer characteristic

Let $V_{GS} = 10\text{ V} \leftarrow$ above $V_{GS(on)}$

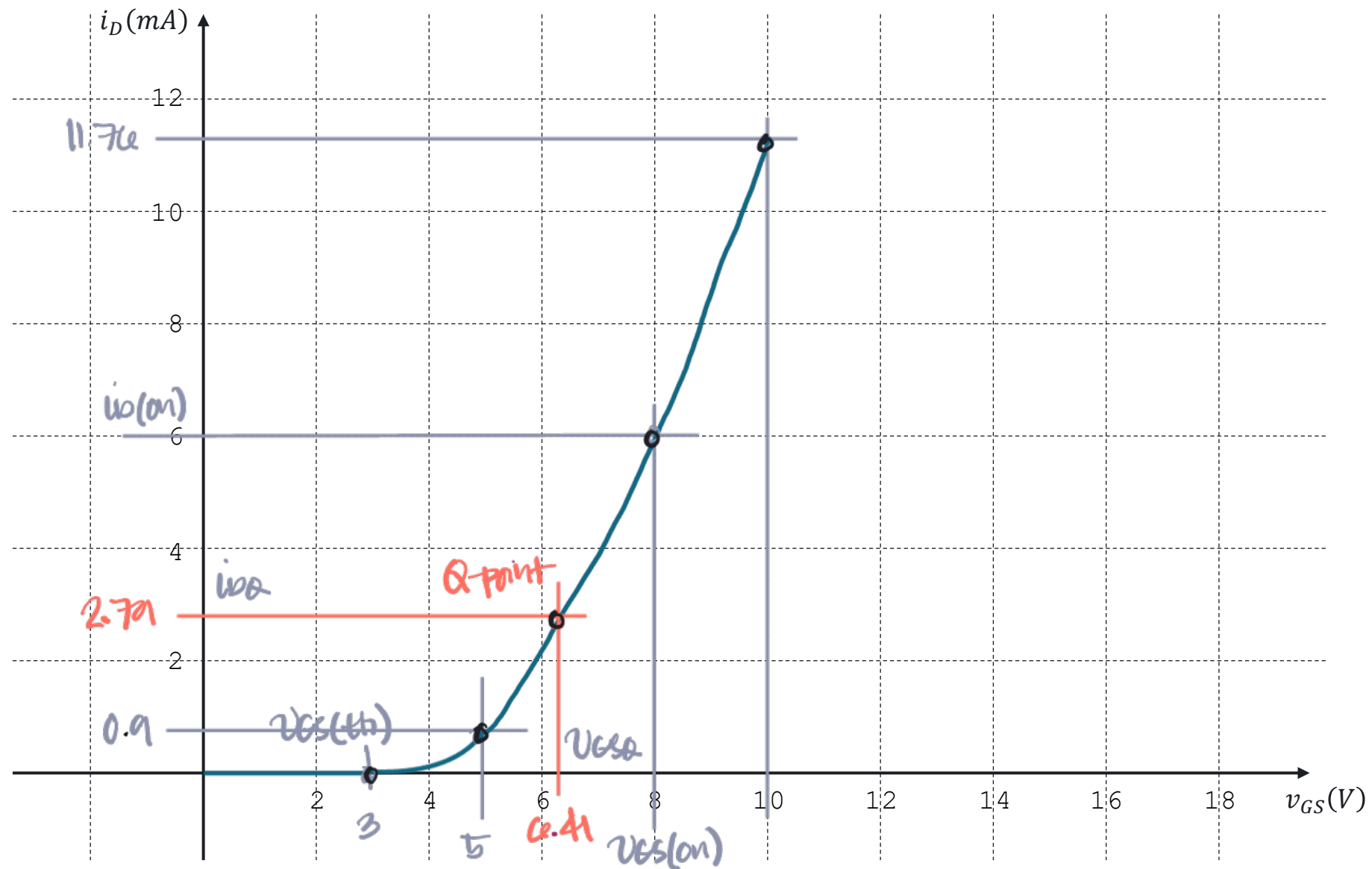
$$i_D = K(V_{GS} - V_{GS(th)})^2$$

$$i_D = 240\mu(10 - 3)^2$$

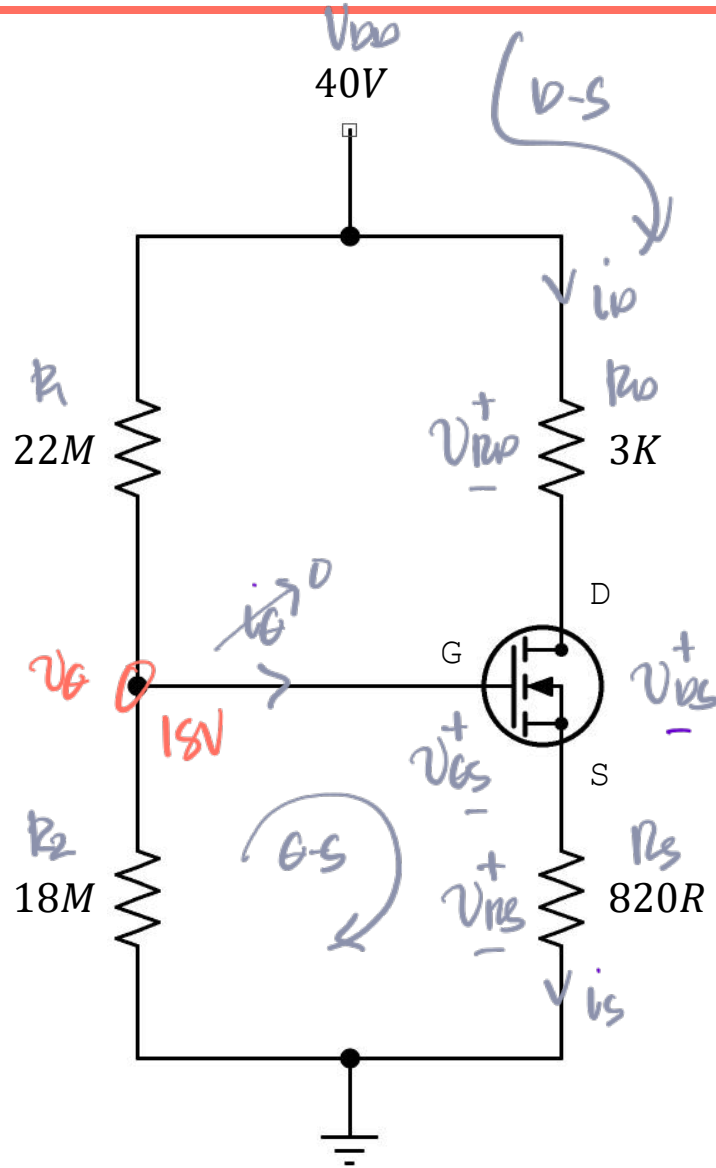
$$i_D = 11.76\text{ mA} \mid V_{GS} = 10\text{ V}$$



TRANSFER CHARACTERISTICS



EXERCISE



$$i_{D(on)} = 3 \text{ mA}$$

$$v_{GS(on)} = 10 \text{ V}$$

$$v_{GS(TH)} = 5 \text{ V}$$

For the given voltage-divider biasing arrangement, determine i_{DQ} , v_{GSQ} and v_{DSQ} for the enhancement-type MOSFET. Sketch the transfer characteristics for $v_{GS} = 8 \text{ V}$ and $v_{GS} = 15 \text{ V}$.

Gate Voltage

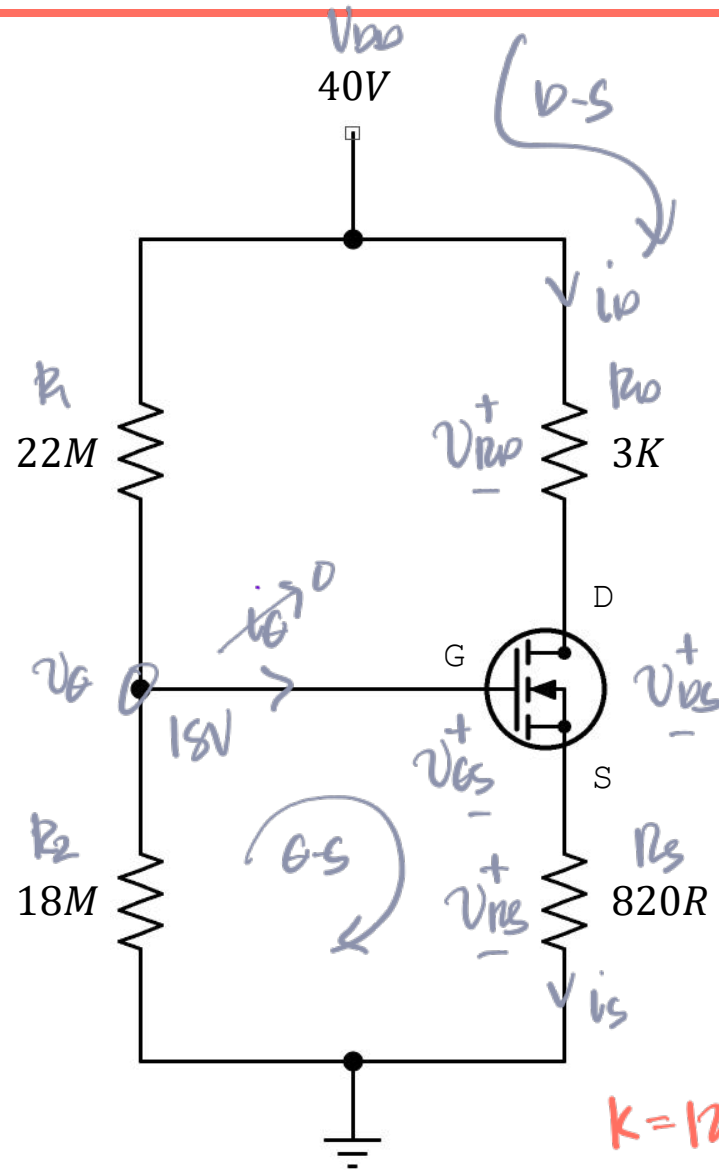
$$V_G = V_{DD} \frac{R_2}{R_1 + R_2}$$

$$V_G = 40 \frac{18\text{M}}{22\text{M} + 18\text{M}}$$

$$\underline{V_G = 18\text{V}}$$



EXERCISE



$$i_{D(on)} = 3 \text{ mA}$$

$$v_{GS(on)} = 10 \text{ V}$$

$$v_{GS(TH)} = 5 \text{ V}$$

Solution

Constant, k

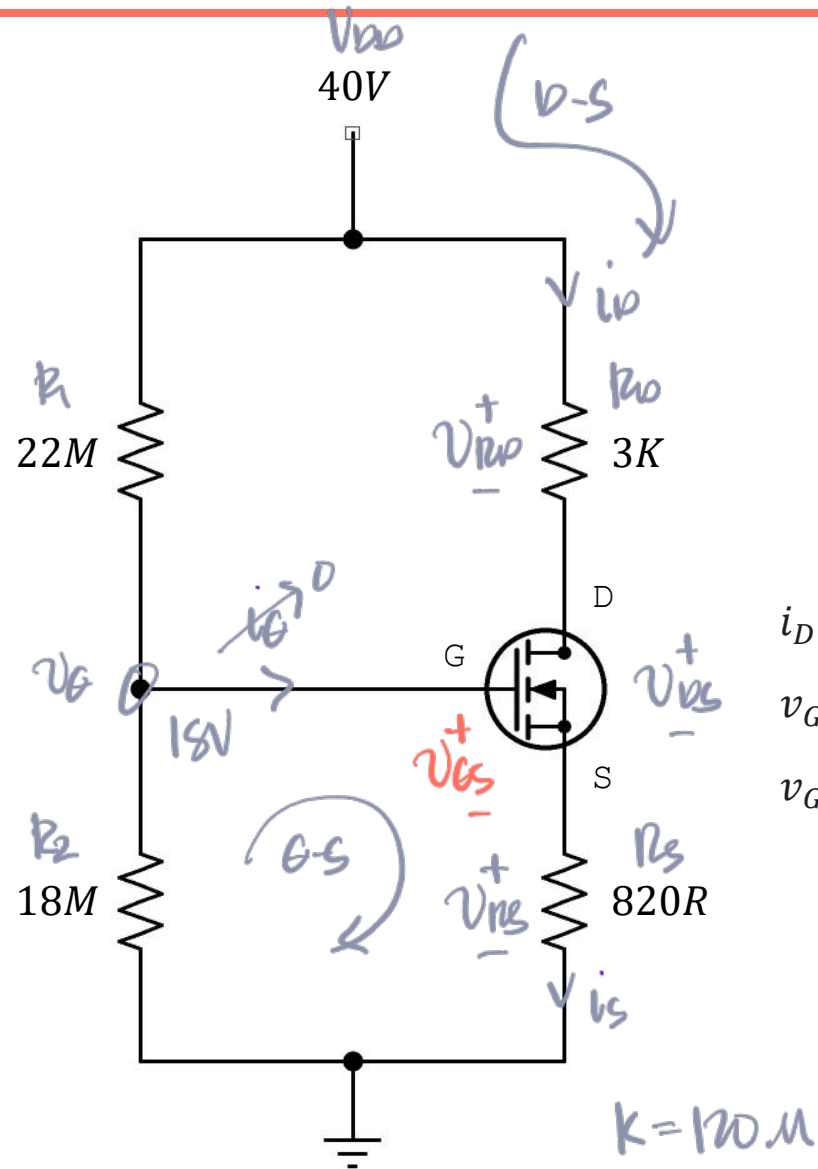
$$k = \frac{i_{D(on)}}{(v_{GS(on)} - v_{GS(th)})^2}$$

$$k = \frac{3 \text{ m}}{(10 - 5)^2}$$

$$\underline{k = 120 \mu\text{A/V}^2}$$



EXERCISE



$$i_{D(on)} = 3 \text{ mA}$$

$$v_{GS(on)} = 10 \text{ V}$$

$$v_{GS(TH)} = 5 \text{ V}$$

Solution

KVL @ G-S

$$-V_G + \cancel{V_{RG}} + V_{GS} + V_{RS} = 0$$

$$V_{GS} = V_G - V_{RS}$$

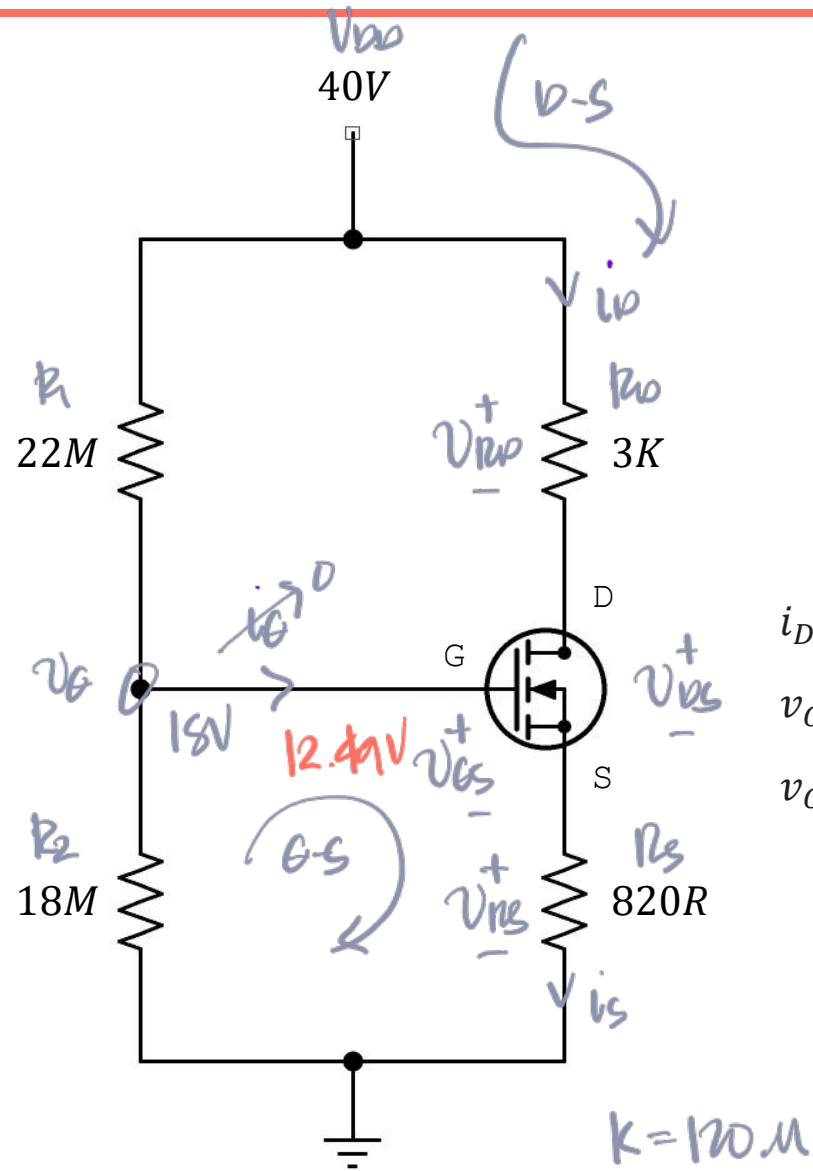
$$V_{GS} = V_G - i_D R_S$$

$$V_{GS} = V_G - K(V_{GS} - V_{GS(TH)})^2 R_S$$

$$V_{GS} = 18 - 120\mu(V_{GS} - 5)^2 \cdot 820$$

$$V_{GS} = 18 - 98.4\mu(V_{GS}^2 - 10V_{GS} + 25)$$

EXERCISE



Solution

KVL @ G-S

$$V_{GS} = 18 - 98.4m V_{GS}^2 + 98.4m V_{GS} - 2.46 - V_{GS}$$

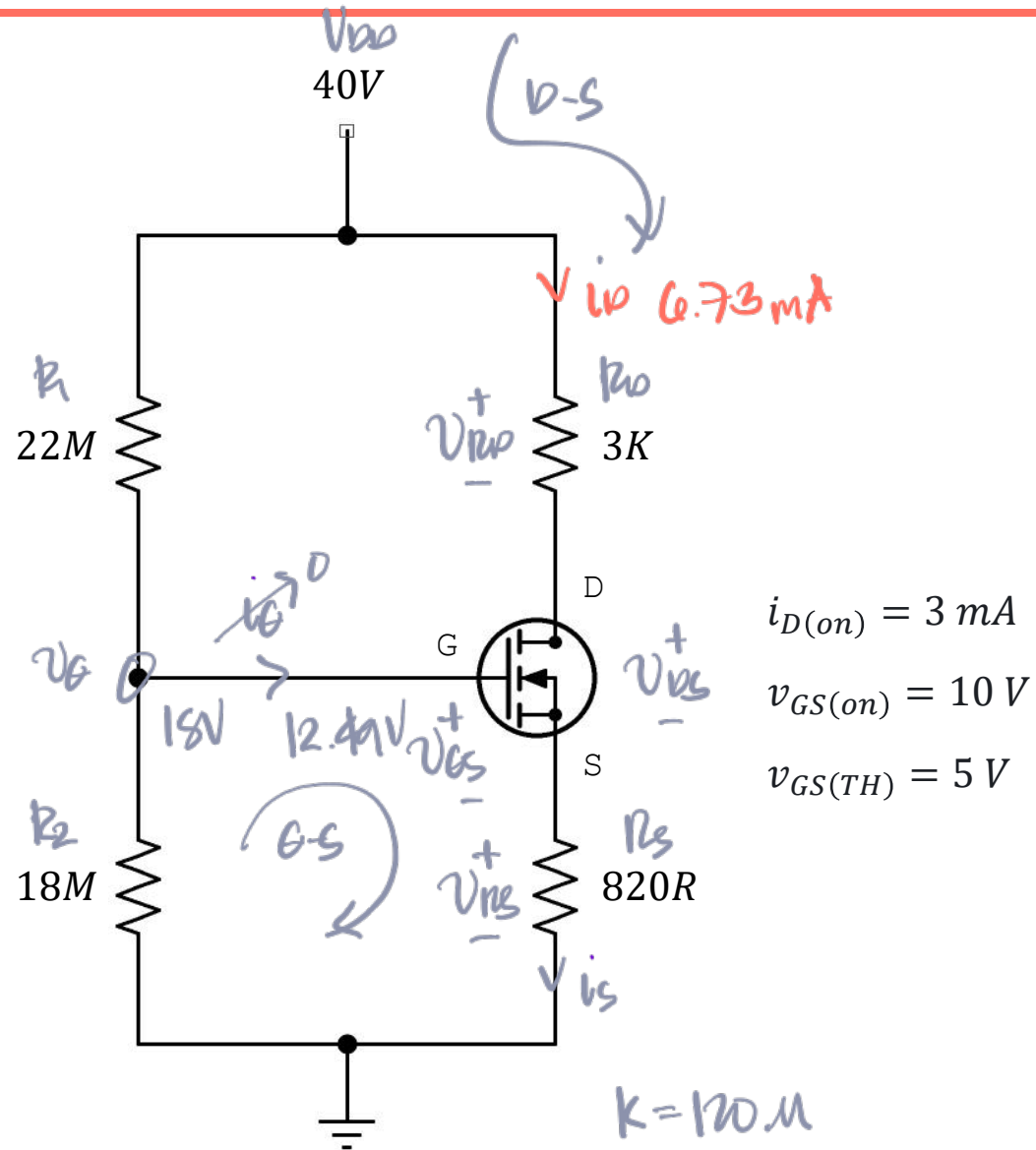
$$0 = -98.4m V_{GS}^2 - 16m V_{GS} + 15.54$$

$$V_{GSQ} = 12.41V$$

ans

$$V_{GS} = -12.65V$$

EXERCISE



Solution

Drain Current

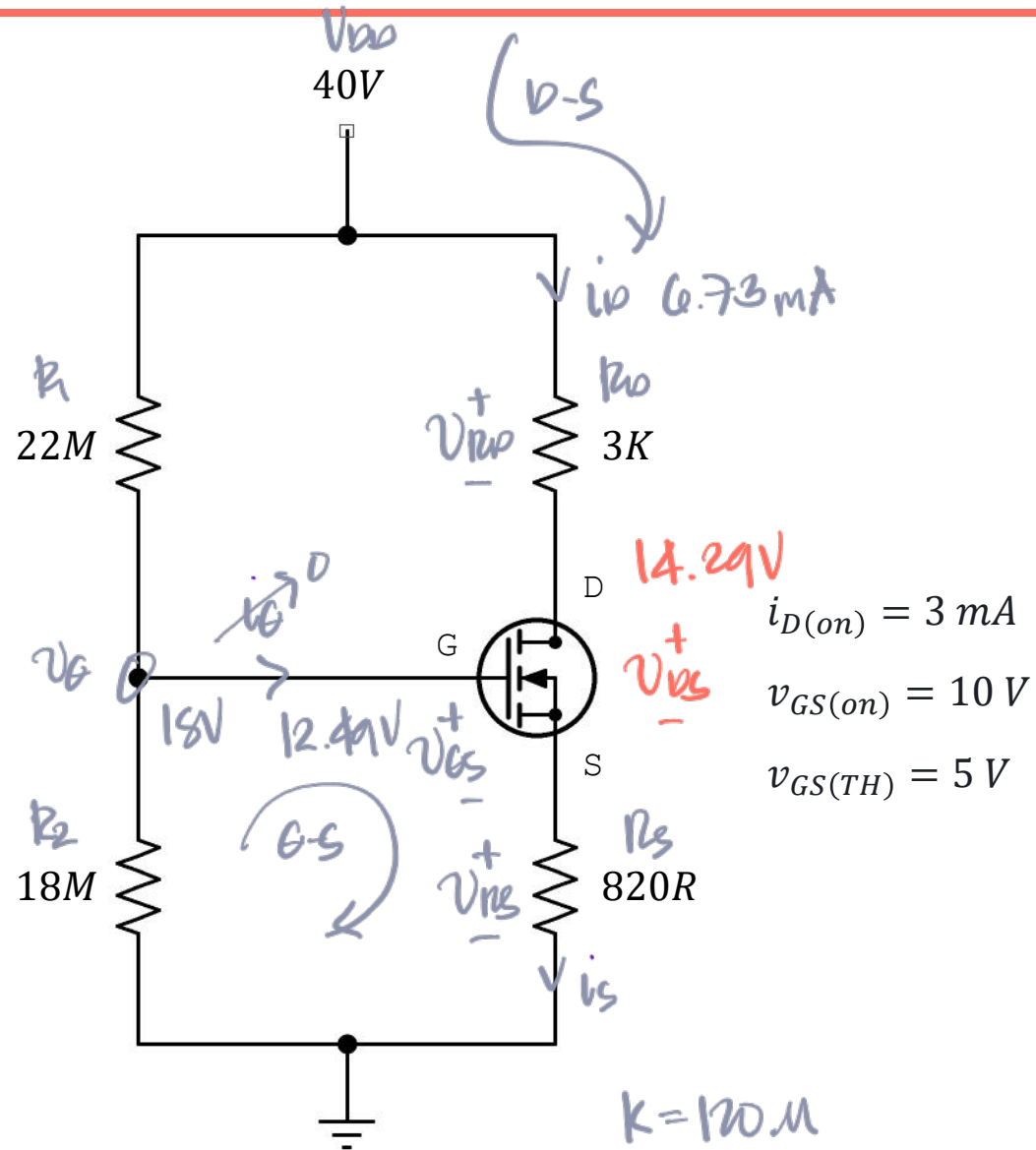
$$i_D = K(V_{GS} - V_{GS(th)})^2$$

$$i_D = 120\mu(12.41 - 5)^2$$

$$i_{DQ} = 6.73mA$$

ans

EXERCISE



Solution

KVL @ D-S

$$-V_{DD} + V_{RD} + V_{DS} + V_{RS} = 0$$

$$V_{DS} = V_{DD} - V_{RD} - V_{RS}$$

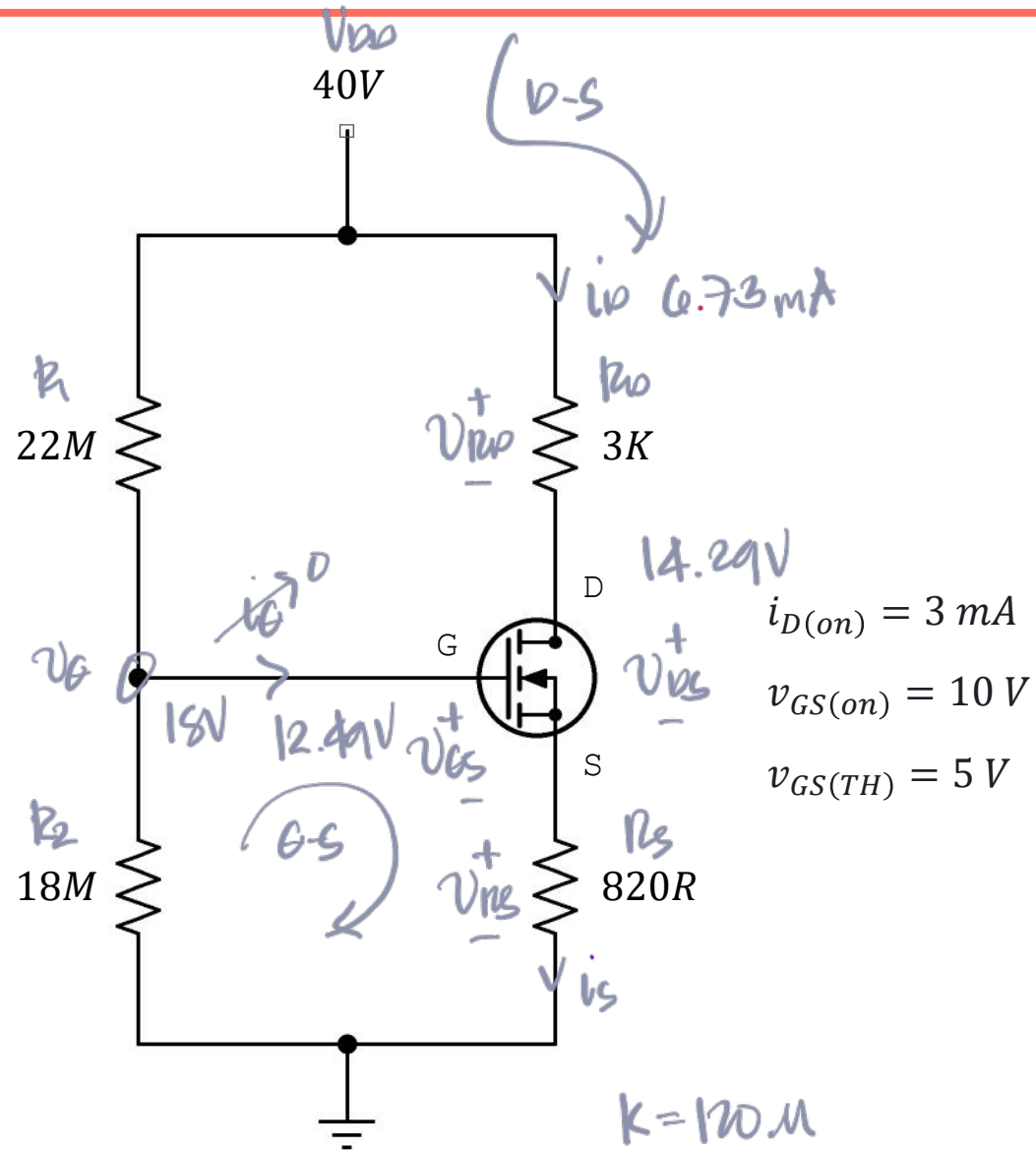
$$V_{DS} = V_{DD} - i_D(R_D + R_S)$$

$$V_{DS} = 40 - 6.73m(3K + 820R)$$

$$V_{DSQ} = 14.29V$$

ans

EXERCISE



Solution

Transfer characteristics

Let $V_{GS} = 8V \leftarrow$ between $V_{GS(th)}^{5V}$ and $V_{GS(on)}^{10V}$

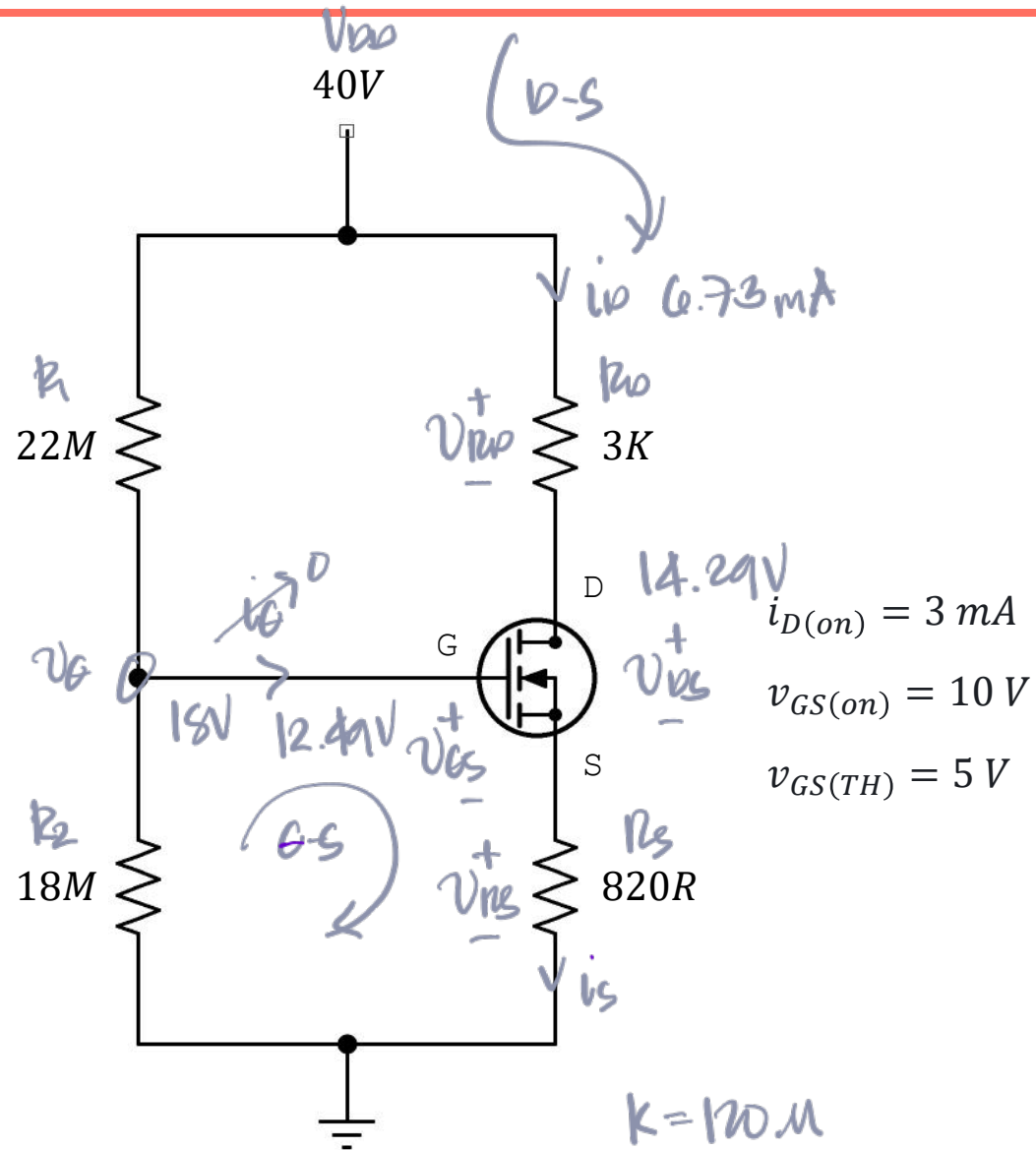
$$i_D = K(V_{GS} - V_{GS(th)})^2$$

$$i_D = 120\mu(8 - 5)^2$$

$$i_D = 1.08mA \mid V_{GS} = 8V$$



EXERCISE



Solution

Transfer characteristics

let $V_{GS} = 15V \leftarrow$ above $V_{GS(on)}$

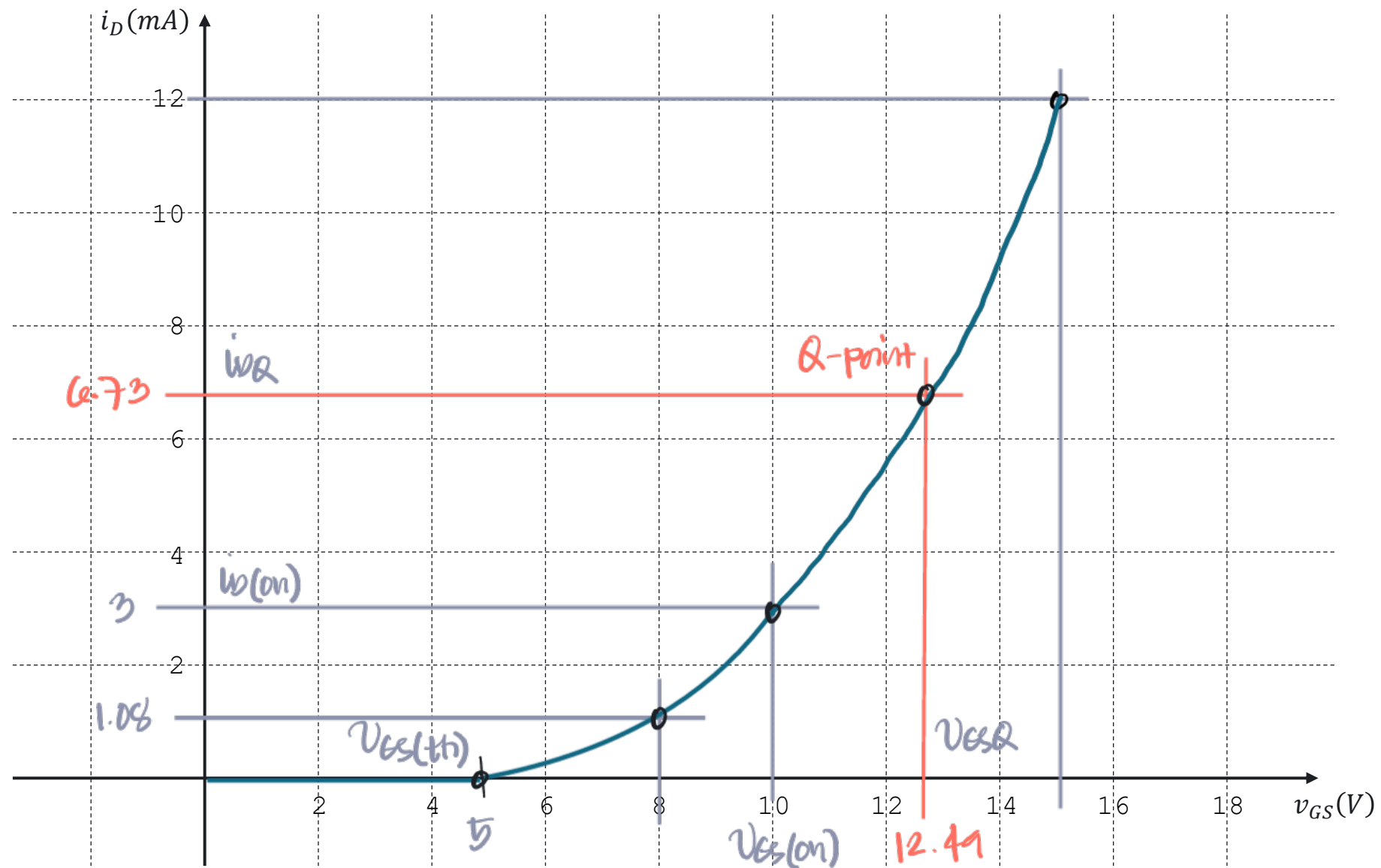
$$i_D = K(V_{GS} - V_{GS(th)})^2$$

$$i_D = 120\mu A (15 - 5)^2$$

$$i_D = 12mA \quad | \quad V_{GS} = 15V$$



TRANSFER CHARACTERISTICS



LABORATORY

